Special Relativity Notes

The Christoffel Symbol and the Covariant Derivative

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1 Covariant Derivative — Motivation

$$\vec{V} = V^i \vec{\epsilon}_i = V^i \left(\frac{\partial x}{\partial q^i} \hat{e}_x + \frac{\partial y}{\partial q^i} \hat{e}_y + \frac{\partial z}{\partial q^i} \hat{e}_z \right)$$

The j-th component of the vector $\vec{V}$ in another coordinate system x^j:

$$(V^i)^j = V^i \frac{\partial x^j}{\partial q^i}$$

Differentiate with respect to q^k:

$$\frac{\partial (V^i)^j}{\partial q^k} = \frac{\partial V^i}{\partial q^k} \frac{\partial x^j}{\partial q^i} + V^i \frac{\partial^2 x^j}{\partial q^k \partial q^i}$$

Because of the extra (second-derivative) term, the derivative of a component of a vector **does not transform as a tensor of rank 2**.

2 The Christoffel Symbol

To fix this, differentiate the vector itself (basis vector and all):

$$\vec{V} = V^i \vec{\epsilon}_i$$

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial V^i}{\partial q^k} \vec{\epsilon}_i + V^i \frac{\partial \vec{\epsilon}_i}{\partial q^k}$$

Since $\vec{\epsilon}_i$ is itself a vector, its derivative can be expanded in the same basis; define the **Christoffel symbol** $\Gamma^j{}_{ik}$ by

$$\frac{\partial \vec{\epsilon}_i}{\partial q^k} = \Gamma^j{}_{ik} \, \vec{\epsilon}_j$$

so that

$$\frac{\partial \vec{V}}{\partial q^k} = \frac{\partial V^i}{\partial q^k} \vec{\epsilon}_i + V^i \, \Gamma^j{}_{ik} \, \vec{\epsilon}_j$$

3 How is Γ Related to the Basis Vectors?

Show that $\Gamma^m{}_{jk} = \vec{\epsilon}^m \cdot \dfrac{\partial \vec{\epsilon}_k}{\partial q^j}$.

Proof. By definition (relabelling the dummy index to n to avoid clashing with m):

$$\frac{\partial \vec{\epsilon}_k}{\partial q^j} = \Gamma^n{}_{jk}\, \vec{\epsilon}_n$$

Dot both sides with $\vec{\epsilon}^m$ and use $\vec{\epsilon}^m \cdot \vec{\epsilon}_n = \delta^m_n$:

$$\vec{\epsilon}^m \cdot \frac{\partial \vec{\epsilon}_k}{\partial q^j} = \Gamma^n{}_{jk}\, (\vec{\epsilon}^m \cdot \vec{\epsilon}_n) = \Gamma^n{}_{jk}\, \delta^m_n = \Gamma^m{}_{jk}$$

$$\boxed{\Gamma^m{}_{jk} = \vec{\epsilon}^m \cdot \frac{\partial \vec{\epsilon}_k}{\partial q^j}}$$

4 Covariant Derivative — Definition

$$\boxed{V^k{}_{;j} = \frac{\partial V^k}{\partial q^j} + \Gamma^k{}_{j\mu}\, V^\mu}$$

The semicolon in $V^k{}_{;j}$ denotes the **covariant derivative** of V^k with respect to q^j — the ordinary derivative $\partial V^k/\partial q^j$, corrected by the Christoffel term $\Gamma^k{}_{j\mu}V^\mu$ so that the result transforms as a genuine rank-2 tensor.